

DB – E(E)R and relational model

Read the descriptions and then answer the following questions:

1. Identify the entities.
2. Identify the relationships between the entities.
3. Determine the cardinalities for each relationship.
4. Identify the attributes present in the entities and in the relationships.
5. Determine the candidate keys and primary keys for each strong entity.
6. Using the answers to (1) – (5):
 - a. Create an ER diagram for the given case study.
 - b. (EER) If appropriate, redraw the diagram by introducing the concepts of specialisation, generalisation, aggregation, and/or composition.
 - c. Represent the database in the form of a Relational ModelExample: Entity(primaryKey, attribute1, attribute2, ...)

I. Film Rental Company

The company has several stores throughout the country. For each store, we intend to store the address (consisting of street, city, district, postal code, and telephone number). Each store can be identified by a unique number.

Employees are assigned to each store, including a manager, who is responsible for the day-to-day management of the store. For each employee, we intend to store the name, position, and salary. Each employee can be identified by a unique number.

Each store holds a stock of films. For each film, we intend to store the catalogue number, video number, title, genre, rental price, cost, status, names of the main actors, and the producer. The catalogue number uniquely identifies each video. However, there are usually several copies of the same video in a store. In this case, individual copies are identified by the video number. Film genres are divided into Action, Drama, Horror, and Science Fiction. The status indicates whether the video is available for rental.

Before being able to rent films, a customer must register as a member. For customers, we intend to store the first and last name, address, and registration date. Each customer is identified by a unique number.

A member may rent up to a maximum of ten films simultaneously. For each rental, we intend to store the rental number; the full name and number of the customer who rented; the number, title, and daily rental price of the video; the rental date; and the return date. The rental number is unique.

II. University

A university has several departments.

For each department, we store a unique identifier, name, building, and telephone number.

Each department employs several lecturers.

For each lecturer, we store a unique identifier, name, academic rank, and salary.

Each department has exactly one head lecturer, who is responsible for managing the department.

The university offers several courses.

For each course, we store a unique course code, title, number of credits, and level (Undergraduate or Postgraduate).

A course belongs to exactly one department.

Each course can have multiple course editions (e.g., different academic years or semesters).

Each edition is identified by a unique edition number and has a semester and academic year.

Students must register before enrolling in courses.

For each student, we store a unique identifier, first name, last name, degree programme, and registration date.

Students may enrol in course editions.

For each enrolment, we store a unique enrolment number, the enrolment date, and the final grade (if available).

A student may enrol in multiple course editions, but only once per edition.

III. Gym

A fitness club manages information about its members, staff, classes, and facilities.

All people associated with the gym are recorded as persons, with attributes such as person ID, name, phone number, and email. A person can either be a member or an employee. Members have additional attributes such as membership start date and status. Employees have attributes such as salary and hire date. Employees are further specialised into trainers (with certification and area of expertise) and administrative staff (with a specific role, such as receptionist or manager).

The gym offers different membership plans. Each plan has a unique ID, name, monthly fee, and access level. Members subscribe to one or more membership plans over time, and for each subscription, the gym records the start date and end date.

The gym organises fitness classes such as yoga, spinning, and strength training. Each class has an ID, name, schedule, and maximum capacity. Classes are led by one or more trainers, and each trainer can lead multiple classes. For each trainer's involvement in a class, the gym records their role (e.g., main instructor, assistant) and the duration of the session.

Members can enrol in classes. A member may enrol in many classes, and each class can have many members. For each enrolment, the gym records the date of enrolment and the attendance status (e.g., attended, missed, cancelled).

The gym also tracks equipment. Each piece of equipment has an ID, name, type, and maintenance status. Trainers may use equipment during classes, and each piece of equipment can be used in multiple classes. For each usage, the gym records the duration of use.

Additionally, members may book personal training sessions. Each session involves one member and one trainer, and has a date, time, and cost. The gym records notes about the session and the member's progress.

IV. Grocery store

The database needs to store products, suppliers, customers, purchases, and employees.

Each product has a unique code, name, price, and current stock level. Products are supplied by different suppliers, and each supplier provides one or more products. For every supplier, the store keeps information such as name, contact details, and address.

Registered customers have a customer ID, name, and contact information, and their purchases are tracked over time.

Each purchase is recorded with a unique identifier, date, and total amount. A sale may include multiple products, and each product can appear in many different sales. The store also records the quantity of each product included in a sale.

Employees work at the store and are responsible for processing sales. Each employee has an ID, name, and role (e.g., cashier, manager). Every sale is processed by a single employee.

V. Vet clinic

A veterinary clinic manages information about people, animals, treatments, and visits.

All people interacting with the clinic are recorded as persons, with attributes such as person ID, name, phone number, and address. A person can either be an owner or an employee. Employees have additional attributes such as salary and hire date. Employees are further specialised into veterinarians (with a licence number and specialisation area) and support staff (with a role such as receptionist or assistant).

Animals (patients) are registered with a unique ID, name, species, breed, and date of birth. Each animal is owned by one owner, but an owner may have multiple animals.

Animals are brought to the clinic for appointments. Each appointment has an ID, date, time, and reason for visit. An appointment involves one animal, but may involve multiple veterinarians, and each veterinarian can participate in many appointments. For each veterinarian's participation in an appointment, the clinic records their role in that visit (e.g., primary vet, assistant) and the time spent.

During appointments, treatments can be performed. Each treatment has a treatment ID, description, and base cost. An appointment may involve multiple treatments, and each treatment can be performed in many appointments. For each treatment performed during an appointment, the clinic records the quantity and the actual cost charged.

The clinic also manages medications. Each medication has an ID, name, and standard dosage. Treatments may require multiple medications, and each medication can be used in multiple treatments. For each medication used in a treatment, the clinic records the dosage and duration.

After each appointment, a bill is generated. Each bill has a unique ID, total amount, and payment status. A bill is associated with one appointment. Owners may make payments toward bills, and a single bill can be paid in multiple payments. Each payment records the date, amount, and payment method.